

The 2a / 2b split: yes, and it does exactly what you want

The identity checks out, the UV lives entirely in 2b — but the infrared log is shared

$$\eta = \frac{p^+}{k^+}, \quad q = k^+ - p^+, \quad \mathcal{C}_{2a}^{af} = -f^{adc} f^{fbe} U^{db}(x) U^{ce}(z) + N_c U^{af}(x), \quad \mathcal{C}_{2b}^{af} = N_c [U^{af}(y) - U^{af}(x)], \quad C_{UV}(\eta) = \frac{\mathcal{P}_{gg}(\eta)}{2N_c}$$

Yes, this is correct, and it is the right split — it repairs exactly the error I flagged last time. $\mathcal{C}_{2a} + \mathcal{C}_{2b}$ reproduces the original colour structure identically (max deviation 2.2×10^{-15}), the index reshuffle $f^{ebf} = -f^{fbe}$ that makes it work is exact, and $\mathcal{C}_{2a}$ vanishes when $U(z) = U(x)$ — so 2b is the collapsed piece and 2a is the genuine three-Wilson-line remainder. **The UV at $z \rightarrow x$ sits entirely in 2b**; 2a is UV finite. One thing to watch: the **large- $|z|$ infrared logarithm is shared** between 2a and 2b, because $\mathcal{C}_{2a} \not\rightarrow 0$ when $U(z) \rightarrow 1$. Plus three placement slips in the LaTeX (§5).

1. The split is an exact identity

The original bracket was $f^{adc} f^{ebf} U^{db}(x) U^{ce}(z) - f^{abc} f^{cbd} U^{df}(y)$. Your two pieces sum to it:

$$\underbrace{-f^{adc} f^{fbe} U^{db}(x) U^{ce}(z) + N_c U^{af}(x)}_{2a} + \underbrace{N_c U^{af}(y) - N_c U^{af}(x)}_{2b} = f^{adc} f^{ebf} U^{db}(x) U^{ce}(z) + N_c U^{af}(y).$$

Two identities do the work, both verified with genuine adjoint $SU(3)$ lines $U^{ab} = 2 \operatorname{tr}(t^a V t^b V^\dagger)$:

- $f^{ebf} = -f^{fbe}$ (swap of the outer indices) — deviation exactly 0. This is what turns your $-f^{adc} f^{fbe}$ into the original's $+f^{adc} f^{ebf}$.
- $-f^{abc} f^{cbd} U^{df}(y) = +N_c U^{af}(y)$ — already checked, 10^{-15} .

Over 40 random configurations: $\max |\mathcal{C}_{2a} + \mathcal{C}_{2b} - \mathcal{C}_{\text{orig}}| = 2.2 \times 10^{-15}$ against $\max |\mathcal{C}_{\text{orig}}| = 3.69$. **Nothing is lost and nothing is added.**

2. 2a vanishes at $U(z) = U(x)$ — and there is a cleaner way to write it

$\|\mathcal{C}_{2a}\|$ with $U(z) = U(x)$ is 1.1×10^{-15} ; with $U(z)$ free it is 0.73. So 2b is precisely the piece that the (illegitimate) collapse would have left, and 2a is precisely what the collapse threw away. That is the right way to organise this.

$$-f^{adc} f^{fbe} U^{db}(x) U^{ce}(z) + N_c U^{af}(x) = -f^{adc} f^{fbe} U^{db}(x) [U^{ce}(z) - U^{ce}(x)]$$

(max deviation 2.2×10^{-15} over 40 configurations.) I would write 2a this way. The $[U(z) - U(x)]$ makes the UV finiteness of §3 visible without any computation, and it makes the physics obvious: 2a only knows about the *difference* between the gluon's Wilson line and the source's.

3. The UV is entirely in 2b

$u^2 \times \langle \text{transverse} \times \text{colour} \rangle_{\hat{u}}$ as $u = z - x \rightarrow 0$; a constant means log divergent, a vanishing entry means finite. Angular average by exact quadrature, with $U(z)$ a smooth adjoint field so that $U(z) \rightarrow U(x)$ properly:

$ z - x $	2b	2a	full row
10^{-2}	0.04786841	1.74×10^{-5}	0.047868404
10^{-3}	0.04786835	1.74×10^{-7}	0.04786831
10^{-4}	0.04786835	1.74×10^{-9}	0.04786835
10^{-5}	0.04786835	1.74×10^{-11}	0.04786835

2a's residue falls as $|z - x|^2$, two powers better than it had to. Two reasons stack:

1. $\mathcal{C}_{2a} = O(|z - x|)$, since $U(z) - U(x) = (z - x) \cdot \partial U + \dots$ (checked: $\|\mathcal{C}_{2a}\|/|z - x|$ is flat at ≈ 1.7 from $|z - x| = 10^{-1}$ down to 10^{-4}). One power gone.

2. That linear piece is **odd** in $\hat{u}$, while the UV residue of the transverse factor is **even** (it came from $\frac{u^m u^n}{u^4}$ structures). The angular average kills the product. A second power gone.

So the whole UV logarithm of the row is $2b$, and since $\mathcal{C}_{2b}$ is z -independent, the residue computed before carries over unchanged:

$$\frac{1}{(2\pi)^3} \frac{g^4}{8\pi^4} \frac{1}{k^+} \int_{\Lambda}^{k^+-\Lambda} \frac{dp^+}{2\pi} \frac{\mathcal{P}_{gg}(\frac{p^+}{k^+})}{2N_c} \int_{x,y,x',w'} \frac{(x'-w') \cdot (y-x)}{(x'-w')^2 (y-x)^2} \left[\int_{UV} \frac{d^2 z}{(z-x)^2} \right] [U(x') - U(w')] N_c [U(y) - U(x)] \rho \rho$$

$\mathcal{P}_{gg}$ times the two LO Weizsäcker–Williams kernels times an LO-like colour structure: this is the piece that renormalises, and it is now cleanly isolated. That is the whole point of the split and it works.

4. But the infrared logarithm is *shared*

At large $|z|$ the gluon's Wilson line goes to the identity, not to $U(x)$, and

$$\mathcal{C}_{2a} \Big|_{U(z) \rightarrow 1} = -f^{adc} f^{fbc} U^{db}(x) + N_c U^{af}(x) \neq 0 \quad (\|\cdot\| = 1.29, \text{ against } N_c \|U\| = 2.43).$$

So $2a$ keeps a large- $|z|$ logarithm too. Numerically, $|z|^2 \times \langle \cdot \rangle$ (constant means log divergent):

$ z $	$2b$	$2a$	$2a + 2b$
10^2	0.0240056	0.0035721	0.0242187
10^3	0.0240333	0.0035762	0.0242468
10^4	0.0240336	0.0035762	0.0242470
10^5	0.0240336	0.0035762	0.0242470

Both are non-zero constants. The practical consequence:

The split is clean for the UV and not for the IR. Treat them differently:

- **UV:** subtract and renormalise on $2b$ alone. $2a$ needs nothing.
- **IR:** the cancellation against the $\mathfrak{B}_2$ (virtual) term of eq. (1) has to be done on $2a$ and $2b$ together. Do not declare $2a$ finite and set it aside — it is UV finite, not finite.

Since $\mathcal{C}_{2a}$ and $\mathcal{C}_{2b}$ have different colour structures at large $|z|$, the two IR pieces will pair with different pieces of $\mathfrak{B}_2$. That is worth checking explicitly before you rely on the cancellation.

5. Three placement slips in the LaTeX (both equations)

1. **The phases are outside the integrals they depend on.** $e^{-ik \cdot (w'-z)} e^{-ik \cdot (z-x)p^+/k^+}$ sits before the $\left\{ \right.$, i.e. outside both $\int_{\Lambda}^{k^+-\Lambda} dp^+$ and $\int_{x,y,z,x'}$ — but it depends on z, x and p^+ . Move it inside, next to the transverse integrand.
2. **The colour structure and the sources are outside the transverse integral.** The first $\left\{ \right.$ closes right after the tensor bracket, leaving $[U^{ab'}(x') - U^{ab'}(w')][\dots]\rho^{b'}(x')\rho^f(y)$ outside $\int_{x,y,z,x'}$ — but they depend on x, y, z, x' . That closing brace belongs at the very end.
3. **Unbalanced braces.** One $\left\{ \right.$ is opened and two $\left\} \right.$ are closed. Fixing (2) fixes this too.

What you *did* fix since last time, and correctly: $\int_{w,w'} \rightarrow \int_{w'}$ in both equations. Still outstanding from before: the $U^{b'a}(w')$ versus $U^{ab'}(w')$ index order in the first bracket, and defining $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$ — the overall sign of the row still rests on them, since the vertex product alone gives a minus.

6. Summary

	colour structure	UV, $z \rightarrow x$	IR, $ z \rightarrow \infty$
2a	$-f^{adc}f^{bde}U^{db}(x)[U^{ce}(z) - U^{ce}(x)]$	finite ($O(z-x ^2)$)	log divergent, residue $\propto 0.0036$ in the test config
2b	$N_c[U^{af}(y) - U^{af}(x)]$	the whole log, residue $C_{UV}(\eta) =$ $\frac{\mathcal{P}_{gg}(\eta)}{2N_c}$	log divergent, residue $\propto 0.0240$

Checks: `src/rowB2_split.py` — §1 the sum identity and $f^{ebf} = -f^{fbe}$; §2 the $[U(z) - U(x)]$ rewriting; §3 the UV scan with a smooth $U(z)$ field; §4 the large- $|z|$ colour limit and scan. Output in `RESULTS_rowB2_split.txt`. The transverse residues themselves are in `src/rowB2_check.py`.